UNIT-1.

SAQ.

1. Define Abstract Data Type.
2. State the main idea behind Bubble sort.
3. Compare and contrast Linear search and Binary search.
4. Differentiate between recursion and iteration.
5. Provide the classification of data structures.

LAQ

1a. Apply the selection sort on the following list of numbers. 21, 11, 5, 78, 49, 54, 72, 88.
 b. Compare different Sorting Techniques.
2a. Write a C program to implement Binary Search Technique.
 b. Explain Linear Search with example.
3a. Apply insertion sort on the following elements. 3,1,4,7,5,9,2,6,5,10.
 b. State and explain Merge sort with an example.
4a. Trace Quick sort algorithm for the following list of numbers. 90,77,60,99,55,88,66.
 b. State and explain Bubble sort with an example.
5a. Discuss the advantage of Binary Search over Linear Search with an example.
 b. Write C program to implement Linear Search

UNIT-2.

SAQ.

1 Define Stack and list out the operations that can be performed on a Stack.
2. Differentiate between Stack and Queue.
3. Define Queue and list out the operations that can be performed on a Queue.
4. Provide any two applications of Queue.
5. Evaluate the postfix expression 2 3 + 4 * 1 2 + 3 / -

LAQ.

1a. Convert the following infix expression into postfix form
A + (B * C) – ((D * E + F) / G)
 b. Describe the steps to evaluate postfix expression
2. Define stack. Explain the operations that can be performed on the stack. Write a C program to implement stack using arrays.
3. Define Queue. . Explain the operations that can be performed on the Queue. Write a C program to implement Queue using arrays.
4. Write the steps to convert infix expression to postfix expression and convert the following infix expression to postfix expression. A * (B + D) / E – F * (G + H / K)
5. Discuss in detail about the following applications of stack.
i) converting infix to postfix expression ii) Evaluating postfix expression

UNIT-3.

SAQ.

1. Differentiate between arrays and linked list.
2. List out different types of linked list.
3. Provide the basic operations carried out on linked list.
4. Write the difference between singly linked list and doubly linked list.
5. List the advantages and disadvantages of doubly linked list over singly linked list.

LAQ.

1a. Discuss in detail about Sparse Matrix and its representations.

b. Explain about the following operations on singly linked list.
i) Deletion at the beginning ii) insertion at end

2. Elucidate in detail about doubly linked list and its operations.

3a. How a Circular linked list is different from singly linked list? Explain.

b. Write short notes on the following operations on a doubly linked list.
i) insertion at the beginning ii) Deletion at end

4a. How to insert a node into Singly linked list? Explain.

b. Differentiate between Singly linked list and doubly linked list.

5. Write a C program that uses functions to perform the following operations on single linked list.(i) Creation (ii) insertion (iii) deletion

UNIT-4.

SAQS

1. Define the terms node, degree ,siblings, height of a Tree.
2. State the properties of a Binary Tree .
3. Write a short note on priority Queue
4. Write about i) Digraph ii) Degree o f a graph
5. List the different graph traversal methods.

LAQS.

1a. Discuss about representation of a Binary Tree .

b. Write inorder , preorder, postorder traversal of the following Tree.

2a. Explain about Breadth First Search Technique with an Example.

b. Write C programs to implement the Depth First Search traversal algorithm

3A. Explain about Depth First Search Technique with an Example.

b. Write C programs to implement the Breadth First Search traversal algorithm

4a. Explain array and linked representation of a Binary Tree.
 b. Demonstrate DFS Technique for the following graph.

5a. Discuss representation of graph with examples.
 b. Describe the concept of Priority Queue.

UNIT-5.

SAQS.

1. List different types of AVL Tree rotations.
2. Provide the methods for resolving collisions.
3. State the properties of AVL Tree.
4. differentiate Binary Search Tree and M-way Search Tree.
5. Give any two application s of Hashing.

LAQS.

1. Describe the concept of hashing and explain how to resolve the collision .
2. Discuss concept of AVL Tree and its rotations with an example.

3a. Construct a Binary Search Tree for the following . 100,50 ,200,25,90,80,150,40,75,30,20.
 b. Write about collision resolving methods.

4. construct B-tree of order 5 with the following elements.
   3,14,7,1,8,5,11,17,13,6,23,12,20,4,16,18,24,25,19

5a. Construct a B-Tree of order 3 with the following elements.
   10,20,15,3,2,16,21,25,30,40
 b. Explain the concept of hashing.